SOLVED PRACTICE WORKBOOK

Biotechnology Fundamentals and DBT Missions - Learner-v2 Refreshed

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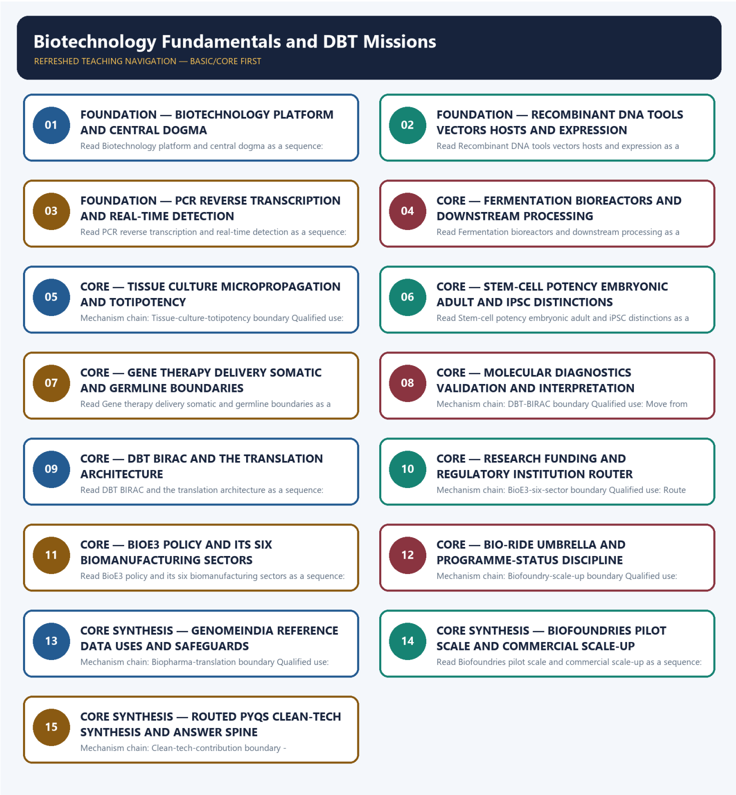

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Biotechnology Fundamentals and DBT Missions - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Biotechnology-platform boundary?

A. Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops. B. Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. C. The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. D. PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis.

Answer: A. Explanation: Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of Biotechnology-platform boundary?

A. PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. B. Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops. C. Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. D. Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene.

Answer: B. Explanation: Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses Biotechnology-platform boundary without changing its institution, unit or status?

A. PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. B. Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. C. Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops. D. Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering.

Answer: C. Explanation: Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA,

diagnostics, culture and fermentation and therefore cannot be reduced to GM crops. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about Biotechnology-platform boundary?

A. Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. B. Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. C. Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. D. Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops.

Answer: D. Explanation: Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Central-dogma boundary?

A. The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. B. PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. C. Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. D. Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene.

Answer: A. Explanation: The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Central-dogma boundary?

A. Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. B. The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. C. PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. D. Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production.

Answer: B. Explanation: The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Central-dogma boundary without changing its institution, unit or status?

A. Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. B. Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. C. The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. D. Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent.

Answer: C. Explanation: The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Central-dogma boundary?

A. Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. B. Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. C. Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. D. The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow.

Answer: D. Explanation: The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies Recombinant-DNA chain?

A. Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. B. Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. C. Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. D. PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis.

Answer: A. Explanation: Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of Recombinant-DNA chain?

A. Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. B. Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. C. Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. D. Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent.

Answer: B. Explanation: Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses Recombinant-DNA chain without changing its institution, unit or status?

A. Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. B. Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. C. Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. D. Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering.

Answer: C. Explanation: Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about Recombinant-DNA chain?

A. Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. B. Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. C. Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. D. Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene.

Answer: D. Explanation: Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies PCR-RT-PCR boundary?

A. PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. B. Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. C. Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. D. Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering.

Answer: A. Explanation: PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of PCR-RT-PCR boundary?

A. Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. B. PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. C. Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. D. Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering.

Answer: B. Explanation: PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses PCR-RT-PCR boundary without changing its institution, unit or status?

A. Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. B. Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. C. PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. D. Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome.

Answer: C. Explanation: PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about PCR-RT-PCR boundary?

A. DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. B. Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. C. Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. D. PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis.

Answer: D. Explanation: PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies Fermentation-bioprocess chain?

A. Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. B. Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. C. Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. D. Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent.

Answer: A. Explanation: Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of Fermentation-bioprocess chain?

A. Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. B. Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. C. Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. D. Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome.

Answer: B. Explanation: Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses Fermentation-bioprocess chain without changing its institution, unit or status?

A. Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. B. DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. C. Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. D. Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome.

Answer: C. Explanation: Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about Fermentation-bioprocess chain?

A. DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable. B. DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. C. Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. D. Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production.

Answer: D. Explanation: Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Tissue-culture-totipotency boundary?

A. Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. B. Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. C. Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. D. Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome.

Answer: A. Explanation: Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Tissue-culture-totipotency boundary?

A. Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. B. Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. C. Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. D. DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator.

Answer: B. Explanation: Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Tissue-culture-totipotency boundary without changing its institution, unit or status?

A. Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. B. DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable. C. Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. D. DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator.

Answer: C. Explanation: Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Tissue-culture-totipotency boundary?

A. DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable. B. BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. C. DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. D. Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering.

Answer: D. Explanation: Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Stem-cell-potency boundary?

A. Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. B. DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. C. Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. D. Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome.

Answer: A. Explanation: Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Stem-cell-potency boundary?

A. DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. B. Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. C. Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. D. DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable.

Answer: B. Explanation: Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Stem-cell-potency boundary without changing its institution, unit or status?

A. DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable. B. DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. C. Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. D. BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research.

Answer: C. Explanation: Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Stem-cell-potency boundary?

A. Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. B. BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. C. DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable. D. Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent.

Answer: D. Explanation: Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies Gene-therapy boundary?

A. Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. B. DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. C. Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. D. DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable.

Answer: A. Explanation: Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of Gene-therapy boundary?

A. DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. B. Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. C. BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. D. DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable.

Answer: B. Explanation: Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses Gene-therapy boundary without changing its institution, unit or status?

A. DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable. B. BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. C. Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. D. Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source.

Answer: C. Explanation: Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about Gene-therapy boundary?

A. Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. B. BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. C. DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties. D. Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome.

Answer: D. Explanation: Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies Molecular-diagnostics boundary?

A. Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. B. DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. C. BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. D. DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable.

Answer: A. Explanation: Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. The other options change the

scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of Molecular-diagnostics boundary?

A. Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. B. Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. C. DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable. D. BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research.

Answer: B. Explanation: Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses Molecular-diagnostics boundary without changing its institution, unit or status?

A. BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. B. Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. C. Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. D. DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties.

Answer: C. Explanation: Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about Molecular-diagnostics boundary?

A. Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. B. DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties. C. A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. D. Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome.

Answer: D. Explanation: Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies DBT-BIRAC boundary?

A. DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. B. Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. C. BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. D. DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable.

Answer: A. Explanation: DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of DBT-BIRAC boundary?

A. BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. B. DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. C. Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. D. DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties.

Answer: B. Explanation: DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses DBT-BIRAC boundary without changing its institution, unit or status?

A. A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. B. Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. C. DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. D. DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties.

Answer: C. Explanation: DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about DBT-BIRAC boundary?

A. A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. B. DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties. C. Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. D. DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator.

Answer: D. Explanation: DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies Institutional-router boundary?

A. DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable. B. DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties. C. BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. D. Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source.

Answer: A. Explanation: DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of Institutional-router boundary?

A. Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. B. DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable. C. A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. D. DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties.

Answer: B. Explanation: DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses Institutional-router boundary without changing its institution, unit or status?

A. A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. B. DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties. C. DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable. D. Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access.

Answer: C. Explanation: DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about Institutional-router boundary?

A. Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. B. A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. C. Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. D. DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable.

Answer: D. Explanation: DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies BioE3-six-sector boundary?

A. BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. B. A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. C. Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. D. DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties.

Answer: A. Explanation: BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of BioE3-six-sector boundary?

A. Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. B. BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. C. DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties. D. A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production.

Answer: B. Explanation: BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses BioE3-six-sector boundary without changing its institution, unit or status?

A. A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. B. Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. C. BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. D. Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale.

Answer: C. Explanation: BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about BioE3-six-sector boundary?

A. Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. B. DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. C. Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. D. BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research.

Answer: D. Explanation: BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies Bio-RIDE boundary?

A. Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. B. A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. C. Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. D. DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties.

Answer: A. Explanation: Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of Bio-RIDE boundary?

A. Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. B. Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. C. A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. D. Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access.

Answer: B. Explanation: Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses Bio-RIDE boundary without changing its institution, unit or status?

A. Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. B. DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. C. Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. D.

Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access.

Answer: C. Explanation: Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about Bio-RIDE boundary?

A. Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. B. DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. C. Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. D. Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source.

Answer: D. Explanation: Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies GenomeIndia boundary?

A. DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties. B. Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. C. A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. D. Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale.

Answer: A. Explanation: DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of GenomeIndia boundary?

A. DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. B. DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties. C. Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. D. Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale.

Answer: B. Explanation: DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses GenomeIndia boundary without changing its institution, unit or status?

A. Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. B. DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. C. DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties. D. Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale.

Answer: C. Explanation: DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about GenomeIndia boundary?

A. DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. B. Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. C. Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. D. DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties.

Answer: D. Explanation: DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Biofoundry-scale-up boundary?

A. A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. B. DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. C. Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. D. Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access.

Answer: A. Explanation: A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of Biofoundry-scale-up boundary?

A. Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. B. A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. C. DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. D. Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards.

Answer: B. Explanation: A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Biofoundry-scale-up boundary without changing its institution, unit or status?

A. DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. B. Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. C. A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. D. Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards.

Answer: C. Explanation: A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Biofoundry-scale-up boundary?

A. Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. B. Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops. C. Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. D. A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production.

Answer: D. Explanation: A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies Biopharma-translation boundary?

A. Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. B. Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. C. Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. D. DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic.

Answer: A. Explanation: Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of Biopharma-translation boundary?

A. Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. B. Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. C. DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. D. Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence.

Answer: B. Explanation: Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses Biopharma-translation boundary without changing its institution, unit or status?

A. Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. B. Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. C. Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. D. Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops.

Answer: C. Explanation: Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about Biopharma-translation boundary?

A. The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. B. Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. C. Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops. D. Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access.

Answer: D. Explanation: Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Microbes-biofuel boundary?

A. Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. B. Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. C. DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. D. Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards.

Answer: A. Explanation: Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Microbes-biofuel boundary?

A. Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. B. Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. C. Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops. D. Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence.

Answer: B. Explanation: Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Microbes-biofuel boundary without changing its institution, unit or status?

A. The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. B. Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. C. Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. D. Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops.

Answer: C. Explanation: Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Microbes-biofuel boundary?

A. Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. B. Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops. C. The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. D. Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale.

Answer: D. Explanation: Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Objective-biotech-concepts boundary?

A. DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. B. Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. C. Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. D. Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops.

Answer: A. Explanation: DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Objective-biotech-concepts boundary?

A. Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. B. DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. C. The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. D. Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops.

Answer: B. Explanation: DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Objective-biotech-concepts boundary without changing its institution, unit or status?

A. Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. B. Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops. C. DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. D. The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow.

Answer: C. Explanation: DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Objective-biotech-concepts boundary?

A. The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. B. Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. C. PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. D. DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic.

Answer: D. Explanation: DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies Clean-tech-contribution boundary?

A. Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. B. Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops. C. Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. D. The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow.

Answer: A. Explanation: Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of Clean-tech-contribution boundary?

A. The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. B. Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. C. Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops. D. Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene.

Answer: B. Explanation: Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses Clean-tech-contribution boundary without changing its institution, unit or status?

A. Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. B. PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. C. Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. D. The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow.

Answer: C. Explanation: Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about Clean-tech-contribution boundary?

A. PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. B. Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. C. Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. D. Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards.

Answer: D. Explanation: Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Status-and-number firewall?

A. Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. B. Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops. C. The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. D. Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene.

Answer: A. Explanation: Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Status-and-number firewall?

A. Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. B. Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. C. PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. D. The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow.

Answer: B. Explanation: Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Status-and-number firewall without changing its institution, unit or status?

A. Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. B. Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. C. Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. D. PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis.

Answer: C. Explanation: Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Status-and-number firewall?

A. Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. B. Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. C. PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. D. Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence.

Answer: D. Explanation: Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Three representative audited cards cover the 2018 and 2021 GS-III biotechnology and biopharma demands; the 2023 microorganisms-and-fuel Mains demand plus routed objective biotechnology concepts; and the 2025 clean-technology biotechnology demand plus provisional 2026 genetic-medicine and GenomeIndia concepts. No objective answer letter or provisional key is invented.

PYQ DEMAND CARD 1 - 2018 and 2021 GS-III

Demand: Discuss biotechnology activity and biopharma benefits in India, and identify applied biotechnology R&D achievements with potential for social upliftment.

Status: Two verified routed Mains demands are consolidated in one representative card. The answer distinguishes scientific achievement from validated, affordable and accessible social outcomes.

Model solution: Biotechnology-platform boundary: Biotechnology uses genes, cells, tissues, enzymes or organisms together with engineering and processing to create knowledge, services or products; it includes recombinant DNA, diagnostics, culture and fermentation and therefore cannot be reduced to GM crops. **Recombinant-DNA chain:** Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. **Fermentation-bioprocess chain:** Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. **DBT-BIRAC boundary:** DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. **Biopharma-translation boundary:** Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. **Status-and-number firewall:** Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 2 - 2023 GS-III and Prelims GS-I

Demand: Explain how microorganisms can address fuel shortages and distinguish routed objective concepts including biofilters, Wolbachia-based vector control and aerial metagenomics.

Status: The verified 2023 Mains fuel demand and routed objective biotechnology concepts are represented together; the unavailable objective answer letters are neither recorded nor inferred.

Model solution: Microbes-biofuel boundary: Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. **Objective-biotech-concepts boundary:** DNA barcoding uses short standard markers with reference databases for identification; aquaculture biofilters rely on attached microbial communities; Wolbachia approaches alter mosquito-vector dynamics; and aerial metagenomics detects airborne DNA without automatically proving that an organism is alive, abundant or pathogenic. **Fermentation-bioprocess chain:** Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. **Molecular-diagnostics boundary:** Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. **Status-and-number firewall:** Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 3 - 2025 and provisional 2026 GS-III and Prelims GS-I

Demand: Assess biotechnology's contribution to clean-technology energy independence and distinguish genetic medicines, delivery vectors, therapeutic DNA modification and GenomeIndia institutional claims.

Status: The verified 2025 Mains demand is combined with the provisional 2026 routed concepts. No provisional option letter or answer key is recorded, inferred or used as evidence for a proposition.

Model solution: Gene-therapy boundary: Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. **GenomeIndia boundary:** DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties. **Clean-tech-contribution boundary:** Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. **Status-and-number firewall:** Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

ORIGINAL MAINS 1 - 10 MARKS

Question: Explain the central dogma and the basic recombinant-DNA workflow. Answer in about 150 words.

Model thesis: Claim: Central-dogma boundary. **Named evidence/example:** The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Recombinant-DNA chain. **Named evidence/example:** Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow.
- Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene.

Qualified conclusion: Claim: Central-dogma boundary. **Named evidence/example:** The core information rail is DNA --transcription--> RNA --translation--> protein; a technique must be located on this rail, while reverse transcription is an explicit RNA-to-DNA operation and not a reversal of every biological information flow. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Recombinant-DNA chain. **Named evidence/example:** Recombinant DNA work joins DNA from different sources through a chain of target identification, restriction-endonuclease cutting, vector insertion, DNA-ligase joining, host transformation, marker selection and expression; a vector is a carrier, not the desired gene. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

ORIGINAL MAINS 2 - 10 MARKS

Question: Differentiate PCR, RT-PCR and molecular diagnostic interpretation. Answer in about 150 words.

Model thesis: Claim: PCR-RT-PCR boundary. **Named evidence/example:** PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Molecular-diagnostics boundary. **Named evidence/example:** Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis.
- Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome.

Qualified conclusion: Claim: PCR-RT-PCR boundary. **Named evidence/example:** PCR amplifies target DNA through denaturation, primer annealing and thermostable-polymerase extension; reverse-transcription PCR first converts RNA to complementary DNA, whereas real-time PCR describes signal measurement during amplification and is a different axis. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Molecular-diagnostics boundary. **Named evidence/example:** Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

ORIGINAL MAINS 3 - 15 MARKS

Question: Examine fermentation, tissue culture and stem-cell platforms as distinct biotechnology tools. Answer in about 250 words.

Model thesis: Claim: Fermentation-bioprocess chain. **Named evidence/example:** Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Tissue-culture-totipotency boundary. **Named evidence/example:** Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Stem-cell-potency boundary. **Named evidence/example:** Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production.
- Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering.
- Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent.

Qualified conclusion: Claim: Fermentation-bioprocess chain. **Named evidence/example:** Industrial bioprocessing controls organism or cell line, nutrients, temperature, pH, oxygen and contamination in a bioreactor, followed by downstream recovery, purification and quality control; fermentation is not confined to alcohol production. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Tissue-culture-totipotency boundary. **Named evidence/example:** Tissue culture grows cells or explants aseptically on controlled media; plant micropropagation rests on totipotency, the capacity of a suitable somatic cell to regenerate a whole plant, but clonal propagation does not itself mean genetic engineering. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Stem-cell-potency boundary. **Named evidence/example:** Stem cells self-renew and differentiate: embryonic stem cells are pluripotent, many adult stem cells are tissue-restricted or multipotent, and induced pluripotent stem cells are reprogrammed somatic cells; pluripotent does not mean totipotent. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

ORIGINAL MAINS 4 - 15 MARKS

Question: Analyse the roles of DBT, BIRAC and sectoral regulators in biotechnology translation. Answer in about 250 words.

Model thesis: Claim: DBT-BIRAC boundary. **Named evidence/example:** DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Institutional-router boundary. **Named evidence/example:** DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biopharma-translation boundary. **Named evidence/example:** Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator.
- DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable.
- Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access.

Qualified conclusion: Claim: DBT-BIRAC boundary. **Named evidence/example:** DBT is the central biotechnology department under the Ministry of Science and Technology for policy, research support and missions, whereas BIRAC is the DBT-set-up not-for-profit Section 8 public sector enterprise and industry-academia translation interface; neither description makes BIRAC a biosafety or drug regulator. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Institutional-router boundary. **Named evidence/example:** DBT, ICMR, CSIR and ICAR support or perform research in different administrative domains, BIRAC supports translation and enterprise, CDSCO regulates drugs and vaccines, and GEAC appraises environmental release of regulated GM organisms; funding, research and approval functions are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biopharma-translation boundary. **Named evidence/example:** Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate BioE3, Bio-RIDE and GenomeIndia as pillars of India's biotechnology strategy. Answer in about 300 words.

Model thesis: Claim: BioE3-six-sector boundary. **Named evidence/example:** BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Bio-RIDE boundary. **Named evidence/example:** Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GenomeIndia boundary. **Named evidence/example:** DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biofoundry-scale-up boundary. **Named evidence/example:** A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Status-and-number firewall. **Named evidence/example:** Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research.
- Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source.
- DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties.
- A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production.
- Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence.

Qualified conclusion: Claim: BioE3-six-sector boundary. **Named evidence/example:** BioE3, approved on 24 August 2024, organises high-performance biomanufacturing around six sectors: bio-based chemicals and enzymes; functional foods and smart proteins; precision biotherapeutics; climate-resilient agriculture; carbon capture and its utilisation; and futuristic marine and space research. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Bio-RIDE boundary. **Named evidence/example:** Bio-RIDE, approved on 19 September 2024, is DBT's umbrella scheme for biotechnology research, innovation, entrepreneurship and a biomanufacturing component; scheme approval is not expenditure, project completion, installed capacity or commercial adoption. The National Biotechnology Development Strategy 2021-2025 is an elapsed historical frame, and no successor may be assumed without a source. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** GenomeIndia boundary. **Named evidence/example:** DBT announced completion of sequencing of 10,000 genomes from the Indian population on 28 February 2024; the resulting population reference can support variant interpretation and research, but a reference dataset is not a clinical diagnosis or a complete map of every Indian community and it retains consent, equity and data duties. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:**

Biofoundry-scale-up boundary. **Named evidence/example:** A biofoundry supports iterative design-build-test-learn work, while pilot and pre-commercial facilities test whether a laboratory process can be controlled and reproduced at larger scale; shared infrastructure access does not establish commercial-scale production. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Status-and-number firewall. **Named evidence/example:** Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

ORIGINAL MAINS 6 - 20 MARKS

Question: Discuss biotechnology's contribution to health, social upliftment and clean-energy security while preserving scale, safety and status boundaries. Answer in about 300 words.

Model thesis: Claim: Gene-therapy boundary. **Named evidence/example:** Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Molecular-diagnostics boundary. **Named evidence/example:** Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biopharma-translation boundary. **Named evidence/example:** Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Microbes-biofuel boundary. **Named evidence/example:** Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Clean-tech-contribution boundary. **Named evidence/example:** Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Status-and-number firewall. **Named evidence/example:** Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome.
- Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome.
- Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access.
- Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale.
- Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards.
- Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence.

Qualified conclusion: Claim: Gene-therapy boundary. **Named evidence/example:** Gene therapy adds, replaces, silences or edits a disease-relevant sequence in selected cells using delivery systems such as engineered viral vectors or lipid nanoparticles; somatic treatment is not heritable germline modification and does not rewrite an entire genome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Molecular-diagnostics boundary. **Named evidence/example:** Molecular diagnostics detect nucleic-acid, protein or other molecular signatures through validated sampling, extraction, amplification or detection and interpretation; detecting a marker is not by itself proof of viable pathogen, disease severity or clinical outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biopharma-translation boundary. **Named evidence/example:** Recombinant proteins, diagnostics, cell systems and bioprocesses can support biopharma and social applications; recombinant human insulin and Ti-plasmid plant transformation are standard mechanism examples, while the National Biopharma Mission and BioNEST illustrate DBT-BIRAC translation support. Research achievement must still pass validation, regulation, manufacturing quality, affordability, procurement and access. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Microbes-biofuel boundary. **Named evidence/example:** Microorganisms can support anaerobic digestion to biogas, fermentation-derived fuels and enzyme-enabled conversion of biomass or waste, but energy value depends on feedstock, land and water use, collection, conversion efficiency, lifecycle emissions and scale. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Clean-tech-contribution boundary. **Named evidence/example:** Biotechnology can complement clean-energy independence through biofuels, waste and biomass conversion, enzymes, bio-based chemicals and carbon-use pathways, but it cannot substitute for renewables, efficiency, storage, grids or lifecycle and ecological safeguards. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Status-and-number firewall. **Named evidence/example:** Discovery, proof of concept, proposal, funded project, pilot, pre-commercial validation, regulatory approval, manufacturing and market adoption are separate rungs; dates, proposal counts, genome counts, capacities, outcomes and objective answer keys require their own dated authoritative evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.